

EmbedUR Linux Training

Module - 5 : Advanced Functions

Assessment - 5

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Question:

Create a Bash script 'file_analyzer.sh', to demonstrate the following concepts:

1. Recursive functions
 - Write a recursive function to search for files in a directory and its subdirectories containing a specific keyword.
2. Redirection and error handling
 - Log errors (e.g., invalid arguments, missing files) to 'errors.log' and display them in the terminal.
3. Here document and here string
 - Use a here document to display a help menu when the '-help' option is passed.
 - Search for a keyword in a specified file using a here string
4. Special parameters
 - Use parameters like '\$0', '\$#', '\$?' and '\$@' to provide meaningful feedback.
5. Regular expressions
 - Validate inputs with regular expressions (Check if the file exists and the keyword is not empty and valid)
6. Command-line arguments using getopt
 - Use 'getopts' to handle:
 - '-d <directory>': Directory to search.
 - '-k <keyword>': Keyword to search.
 - '-f <file>': File to search directly.
 - '-help': Display the help menu.

Example usage:

```
# Recursively search a directory for a keyword
```

```
./file_analyzer.sh -d logs -k error
```

```
# Search for a keyword in a file
```

```
./file_analyzer.sh -f script.sh -k TODO
```

```
# Display the help menu
```

```
./file_analyzer.sh --help
```

Solution:

```
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Assessments/Module - 5
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Assessments/Module - 5$ ls -l
total 52
-rw-rw-r-- 1 kishore kishore  0 Feb  9 08:37  errors.log
-rw-rw-r-- 1 kishore kishore 2446 Feb  9 08:36  file_analyzer.sh
-rw-rw-r-- 1 kishore kishore 42572 Feb  8 11:19 'Here Document Illustration.png'
-rw-rw-r-- 1 kishore kishore 1146 Feb  9 08:31  Recursive_File_Operations.sh
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Assessments/Module - 5$ cat file_analyzer.sh
#!/bin/bash

ERROR_LOG="errors.log"
: > "$ERROR_LOG" # clear old errors

# ----- HELP MENU (HERE DOCUMENT) -----
show_help() {
cat << EOF
Usage: $0 [OPTIONS]

Options:
-d <directory>  Directory to search recursively
-k <keyword>    Keyword to search for
-f <file>       File to search directly
--help         Display this help menu

Examples:
$0 -d logs -k error
$0 -f script.sh -k TODO
$0 --help
EOF
}

# ----- ERROR HANDLER -----
error() {
    echo "ERROR: $1" | tee -a "$ERROR_LOG" >&2
}

# ----- REGEX VALIDATION -----
valid_keyword() {
    [[ "$1" =~ [^[:space:]] ]]
}

# ----- RECURSIVE FUNCTION -----
search_recursive() {
    local dir="$1"
    local keyword="$2"

    for item in "$dir"/*; do
        [[ ! -e "$item" ]] && continue
    done
}
```

```
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        elif [[ -f "$item" ]]; then
            if grep -q "$keyword" "$item" 2>>"$ERROR_LOG"; then
                echo "Found in: $item"
            fi
        fi
    done
}

# ----- GETOPTS -----
while getopts ":d:k:f:~" opt; do
    case "$opt" in
        d) DIRECTORY="$OPTARG" ;;
        k) KEYWORD="$OPTARG" ;;
        f) FILE="$OPTARG" ;;
        ~)
            [[ "$OPTARG" == "help" ]] && show_help && exit 0
            ;;
        *)
            error "Invalid option"
            show_help
            exit 1
            ;;
    esac
done

# ----- ARGUMENT CHECK -----
if [[ $# -eq 0 ]]; then
    error "No arguments provided"
    show_help
    exit 1
fi

# ----- KEYWORD VALIDATION -----
if ! valid_keyword "$KEYWORD"; then
    error "Invalid or empty keyword"
    exit 1
fi

# ----- DIRECTORY SEARCH -----
if [[ -n "$DIRECTORY" ]]; then
    if [[ ! -d "$DIRECTORY" ]]; then
        error "Directory does not exist: $DIRECTORY"
        exit 1
    fi
    search_recursive "$DIRECTORY" "$KEYWORD"
fi

# ----- FILE SEARCH (HERE STRING) -----
if [[ -n "$FILE" ]]; then
    if [[ ! -f "$FILE" ]]; then
        error "File does not exist: $FILE"
        exit 1
    fi
    if grep -q "$KEYWORD" "$FILE" 2>>"$ERROR_LOG"; then
        echo "Found in: $FILE"
    fi
fi
```

```
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;;
esac
done
# ----- ARGUMENT CHECK -----
if [[ $# -eq 0 ]]; then
    error "No arguments provided"
    show_help
    exit 1
fi
# ----- KEYWORD VALIDATION -----
if ! valid_keyword "SKEYWORD"; then
    error "Invalid or empty keyword"
    exit 1
fi
# ----- DIRECTORY SEARCH -----
if [[ -n "$DIRECTORY" ]]; then
    if [[ ! -d "$DIRECTORY" ]]; then
        error "Directory does not exist: $DIRECTORY"
        exit 1
    fi
    search_recursive "$DIRECTORY" "$KEYWORD"
fi
# ----- FILE SEARCH (HERE STRING) -----
if [[ -n "$FILE" ]]; then
    if [[ ! -f "$FILE" ]]; then
        error "File does not exist: $FILE"
        exit 1
    fi

    if grep -q "$KEYWORD" <<< "${cat "$FILE"}"; then
        echo "Keyword found in file: $FILE"
    else
        echo "Keyword not found in file: $FILE"
    fi
fi
# ----- SPECIAL PARAMETERS -----
echo "Script Name      : $0"
echo "Arguments Count   : $# "
echo "Arguments Provided : $@"
echo "Last Exit Status   : $?"

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```

Output:

```
Activities Terminal Feb 9 08:37
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Assessments/Module - 5
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Assessments/Module - 5$ bash file_analyzer.sh -d "/home/kishore/Desktop/Linux Training" -k Log
Found in: /home/kishore/Desktop/Linux Training/Linux Technical Session/Campus Hires 2025 Batch _ Linux Training Program _ Technical Session 1-20250120_110613-Meeting Recording.mp4
Found in: /home/kishore/Desktop/Linux Training/Linux Technical Session/Campus Hires 2025 Batch _ Linux Training Program _ Technical Session 2-20250123_105859-Meeting Recording.mp4
Found in: /home/kishore/Desktop/Linux Training/Linux Technical Session/Campus Hires 2025 Batch _ Linux Training Program _ Technical Session 3-20250127_110037-Meeting Recording.mp4
Found in: /home/kishore/Desktop/Linux Training/Linux Technical Session/Campus Hires 2025 Batch _ Linux Training Program _ Technical Session 6 (part 1)-20250205_110441-Meeting Recording.mp4
Found in: /home/kishore/Desktop/Linux Training/Linux Technical Session/Campus Hires 2025 Batch _ Linux Training Program _ Technical Session 6 (part 2)-20250210_110404-Meeting Recording.mp4
Script Name      : file_analyzer.sh
Arguments Count   : 4
Arguments Provided : -d /home/kishore/Desktop/Linux Training -k Log
Last Exit Status : 0
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Assessments/Module - 5$ bash file_analyzer.sh -f "/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Log.txt" -k backup
Keyword found in file: /home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Log.txt
Script Name      : file_analyzer.sh
Arguments Count   : 4
Arguments Provided : -f /home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Log.txt -k backup
Last Exit Status : 0
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Assessments/Module - 5$ bash file_analyzer.sh --help
Usage: file_analyzer.sh [OPTIONS]

Options:
  -d <directory>  Directory to search recursively
  -k <keyword>    Keyword to search for
  -f <file>       File to search directly
  --help          Display this help menu

Examples:
  file_analyzer.sh -d logs -k error
  file_analyzer.sh -f script.sh -k TODO
  file_analyzer.sh --help
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Assessments/Module - 5$
```